

CS & IT ENGINEERING

DISCRETE MATHS
SET THEORY



Lecture No. 12



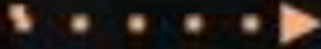
By- SATISH YADAV SIR

A stylized laptop with a blue screen and an orange base. The screen displays the text 'TOPICS TO BE COVERED' in a black serif font, enclosed in a white rectangular box with a blue border.

TOPICS TO BE COVERED

A horizontal dotted orange arrow pointing to the right.

01 Basics of relations

A horizontal dotted orange arrow pointing to the right.

02 Types of relations

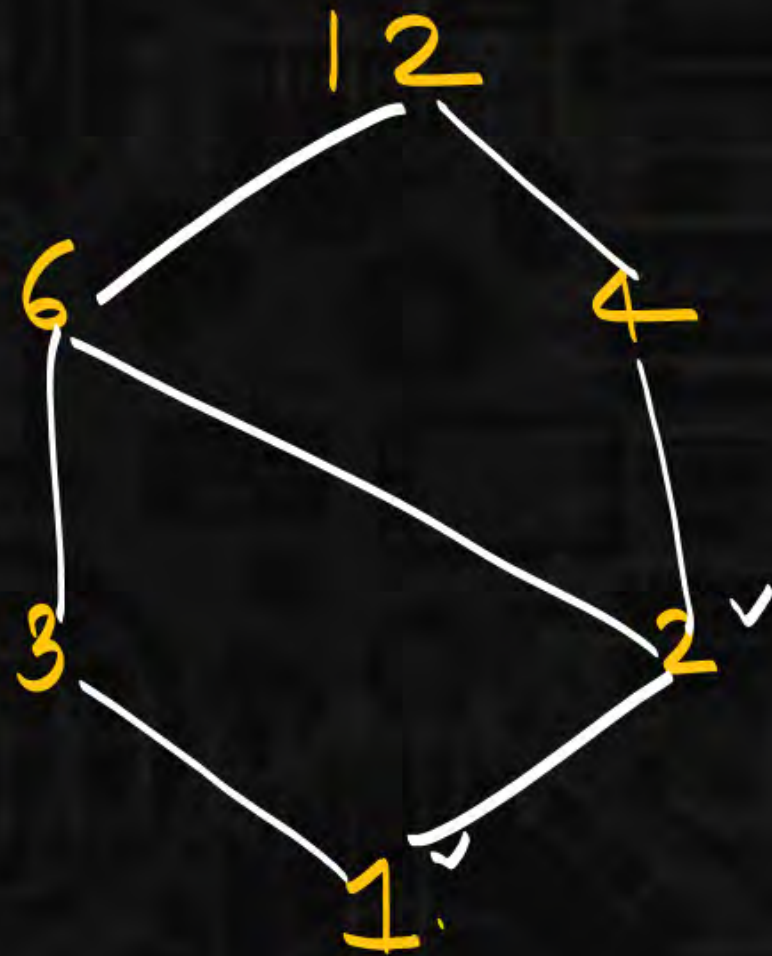
A horizontal dotted orange arrow pointing to the right.
A horizontal dotted orange arrow pointing to the right.

03 Number Of relations

Relations



$(\underline{D|2}, 1)$ poset.
 $(\{1, 2, 3, 4, 6, 12\}, |)$



$$\text{lub}(2, 3) = 6$$

$$\text{lcm}(2, 3) = 6$$

$$\text{ub}(2, 3) = \{6, 12\}$$

$$\text{lub}(2, 3) = \{6\}$$

$$\underline{\text{lub}(6, 4) = \{12\}} \quad \text{lcm}(6, 4) = 12$$

$$\underline{\text{glb}(6, 4) = \{2\}}$$

$$\text{gcd}(6, 4) = 2$$

Relations

$q|b \rightarrow a|c|d$
 $l|u|b \rightarrow l|c|m$

$(\underline{D_n}, 1)$



$(D_{24}, 1)$

$(\{1, 2, 3, 4, 6, 8, 12, 24\}, 1)$

$q|b(6, 8) = 2$

$q|c|d(6, 8) = 2$

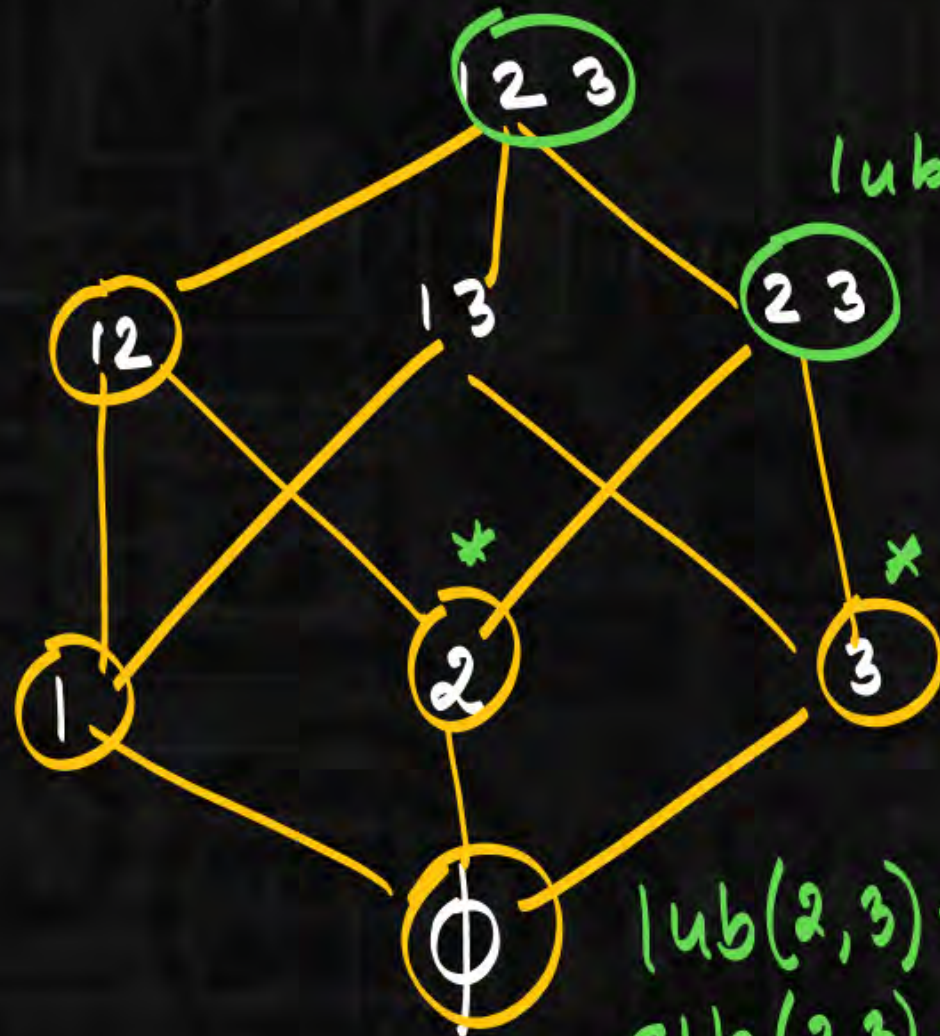
Relations

$$A = \{1, 2, 3\}$$

$(P(A), \subseteq)$ poset

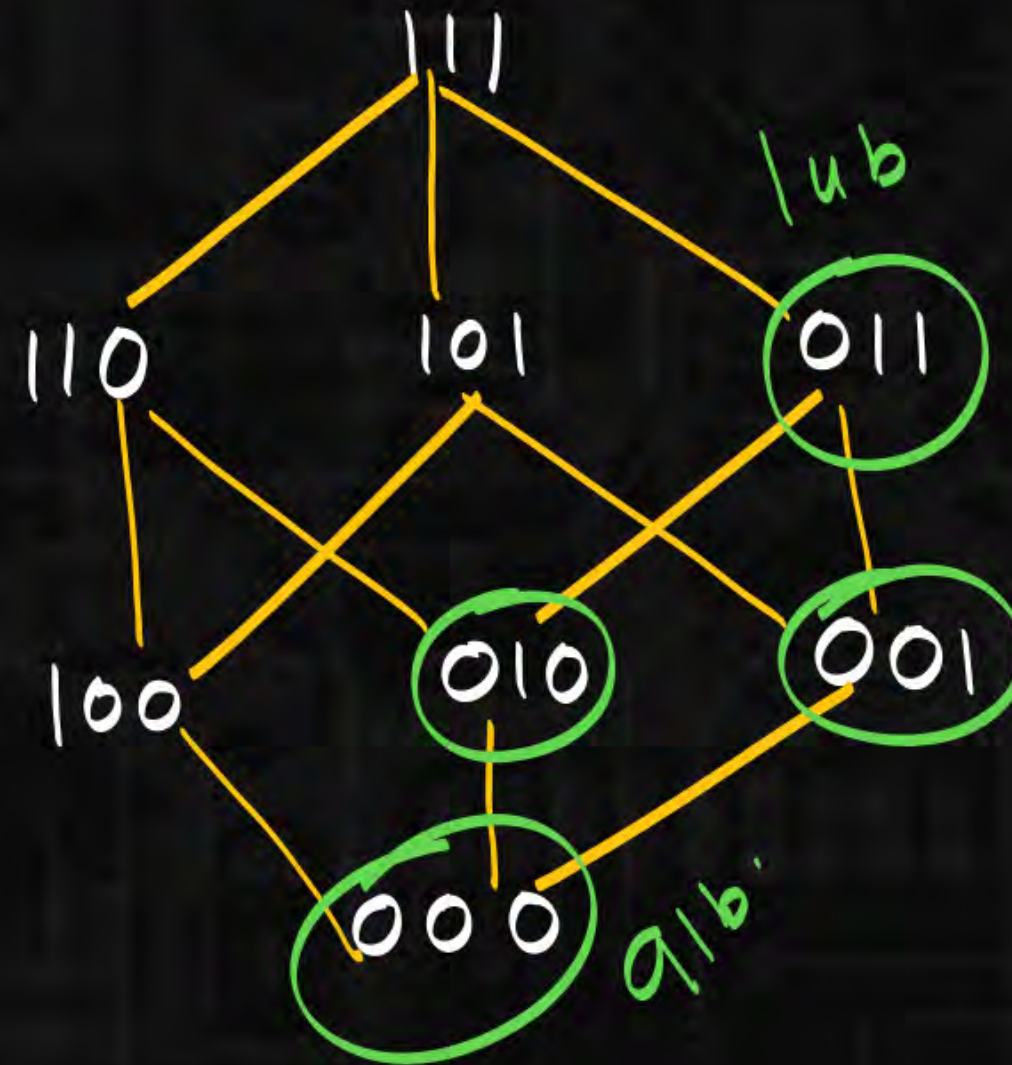
$$\text{lub}(2, 3) = \{2, 3\} \rightarrow U$$

$$\text{glb}(2, 3) = \emptyset \rightarrow n$$



$$\text{lub}(2, 3) = \{2, 3\}$$

$$\text{glb}(2, 3) = \emptyset$$



$$\begin{array}{r} 010 \\ + 001 \\ \hline 011 \end{array} \quad \text{lub} \rightarrow +$$

$$\text{lub}(010, 001) = \underline{\underline{011}}$$

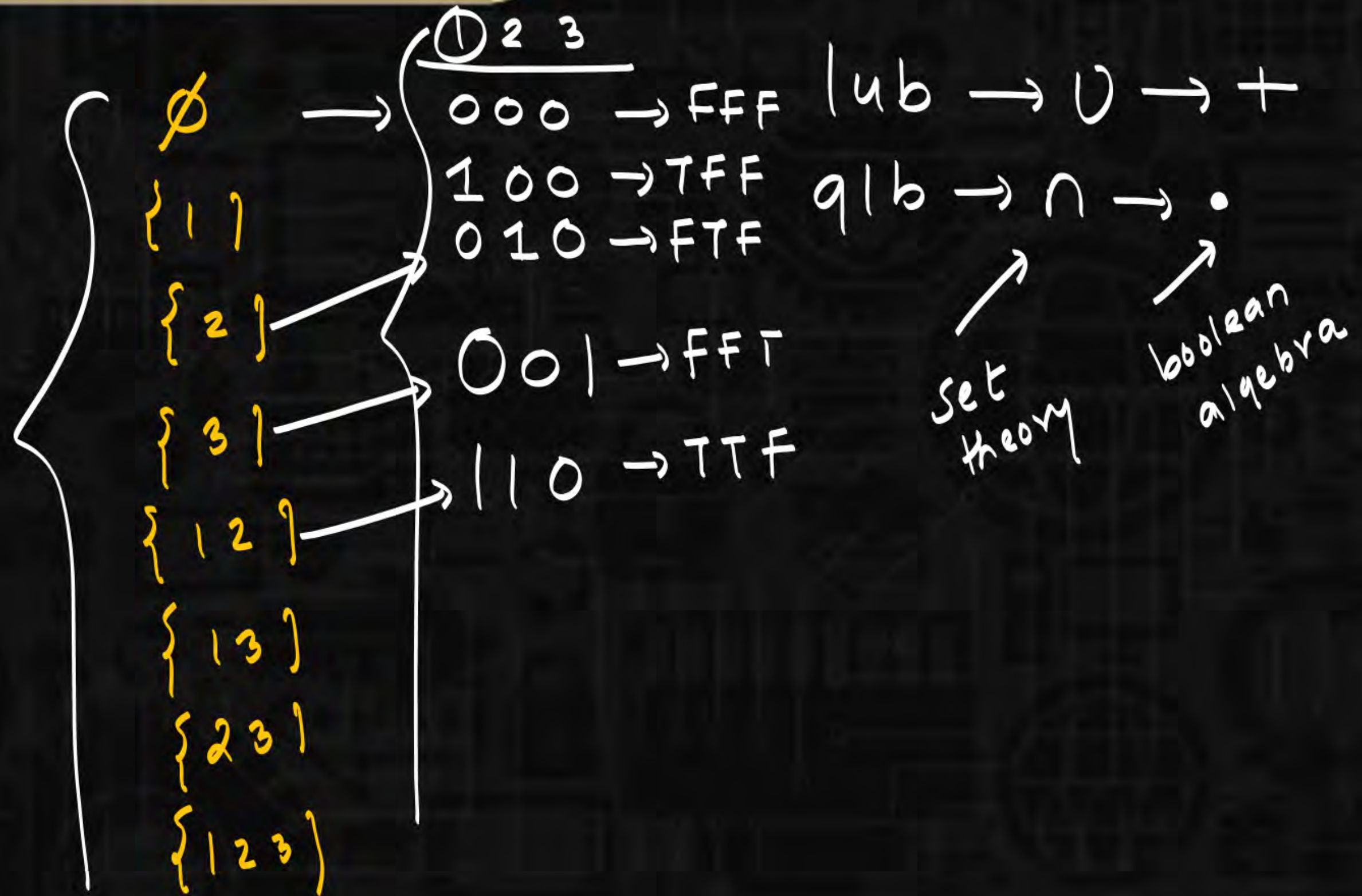
$$\text{glb} \rightarrow \cdot$$

$$\text{glb}(010, 001) = 000$$

$$\begin{array}{r} 010 \\ \cdot 001 \\ \hline 000 \end{array}$$

Relations

$0 \rightarrow f$ $1 \rightarrow T$



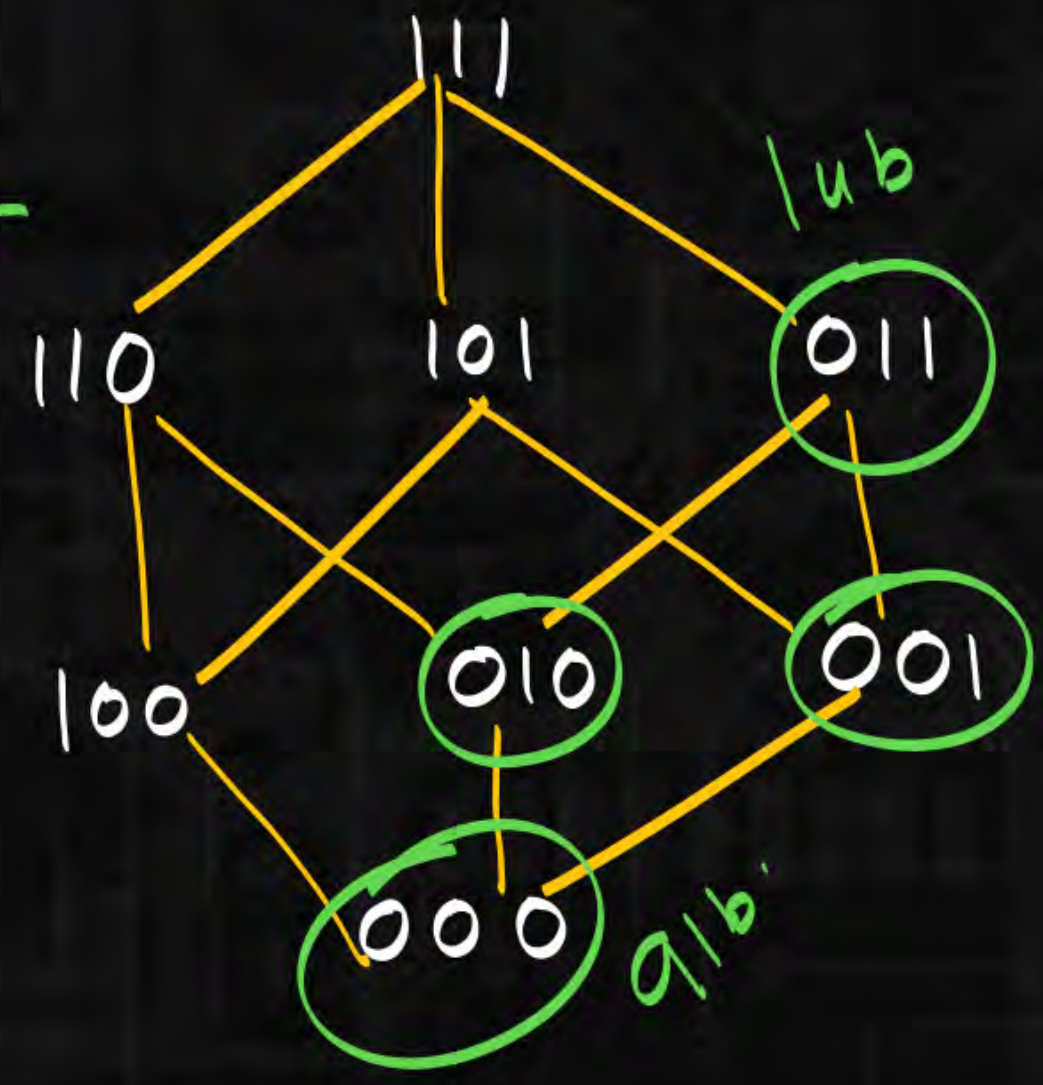
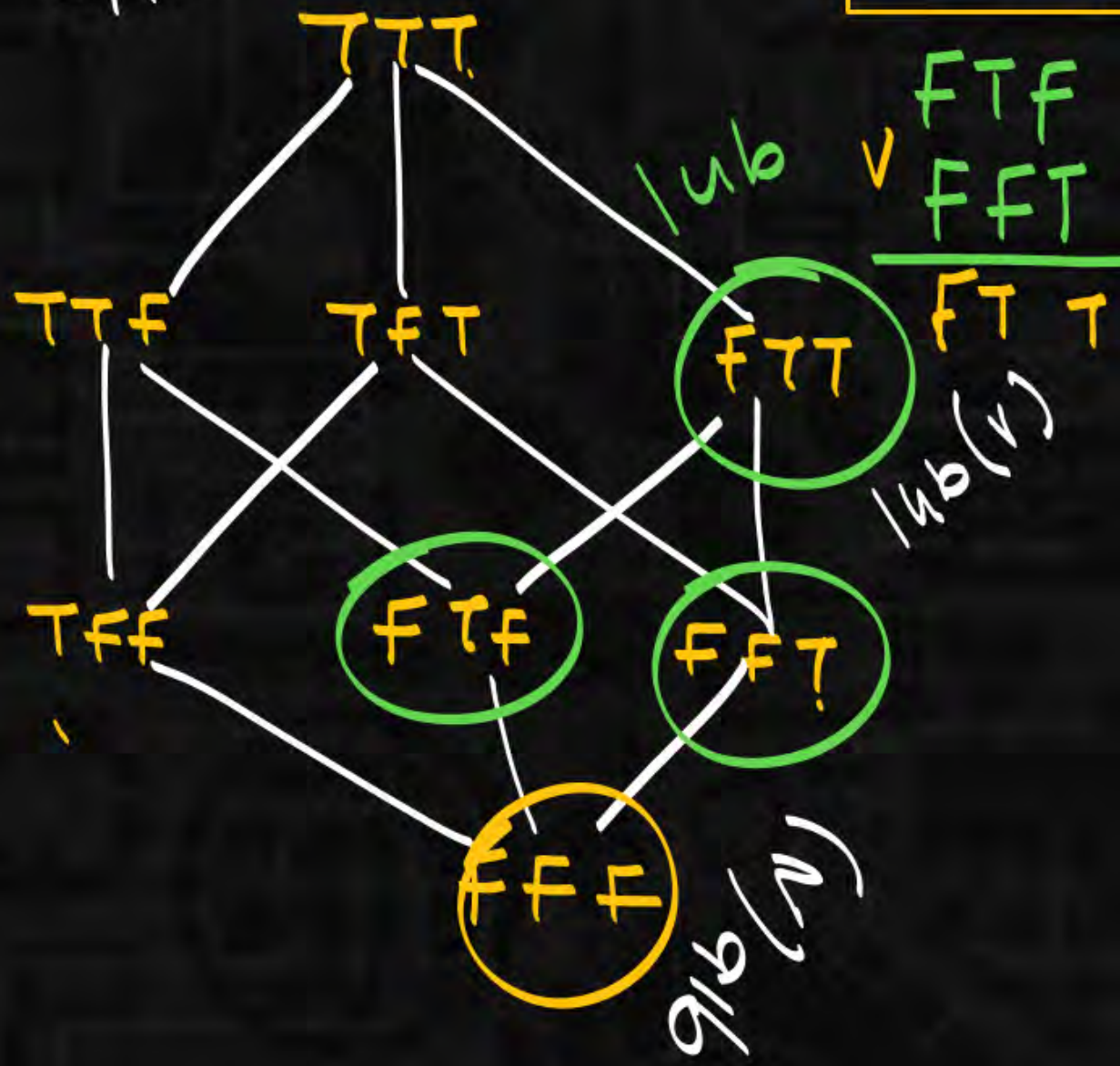
Relations

$\text{lub} \rightarrow \vee$
 $\text{gib} \rightarrow \wedge$

$$\begin{aligned} \text{lub}(FTF, FFT) &= FTT \\ \text{gib}(FTF, FFT) &= FFF \end{aligned}$$

$$\begin{array}{ccc} \vee & F & T & F \\ & F & F & T \\ \hline & F & T & T \end{array}$$

$$\begin{array}{ccc} \wedge & F & T & F \\ & F & F & T \\ \hline & F & F & F \end{array}$$



Relations

(D_n, l)

lcm

gcd

$lub \rightarrow \cup \rightarrow \vee \rightarrow +$

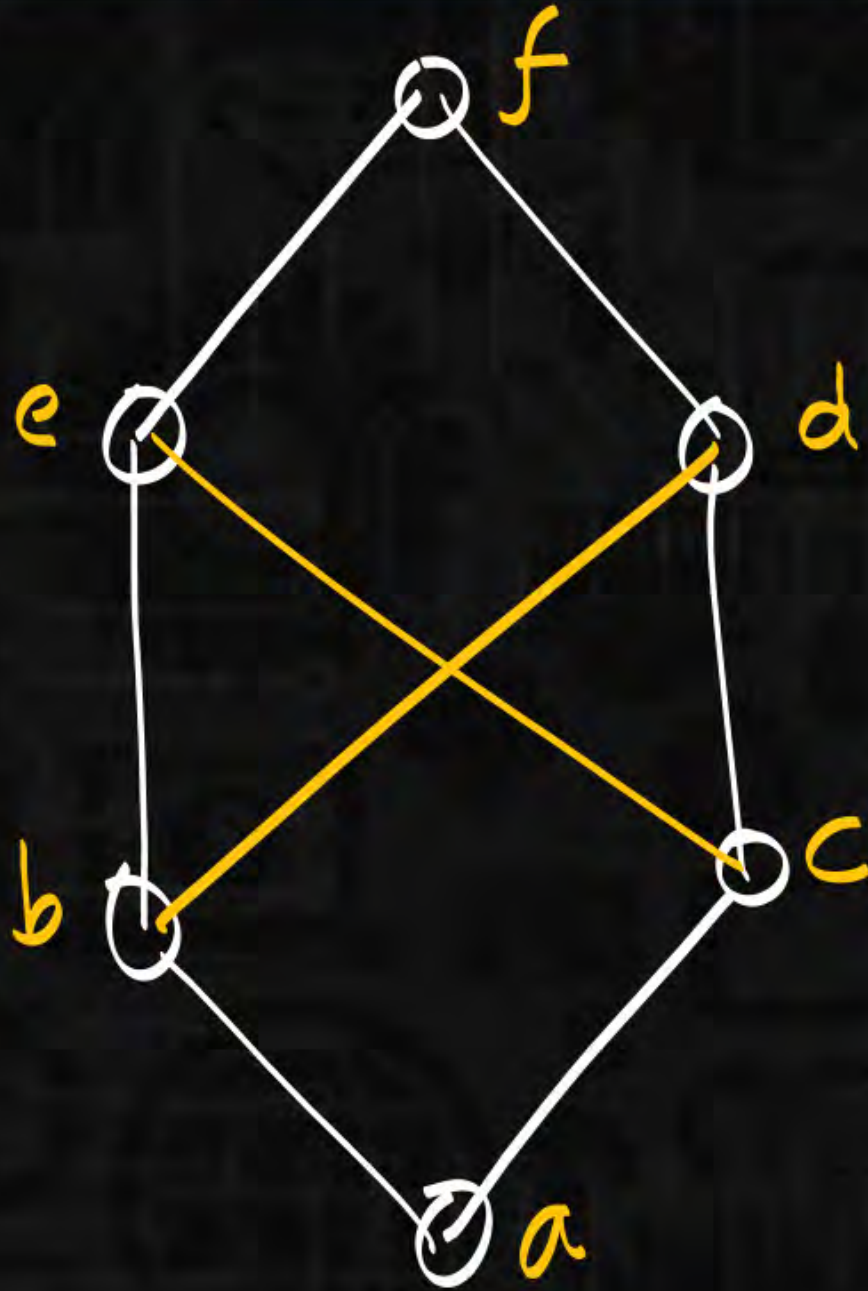
$glb \rightarrow \cap \rightarrow \wedge \rightarrow \cdot$

set theory.

logic

b.A

Relations



$$\text{lub}(e, d) = f$$

$$\text{glb}(e, d) = \text{N.A.}$$

$$\text{lb}(e, d) = \{b, c, a\}$$

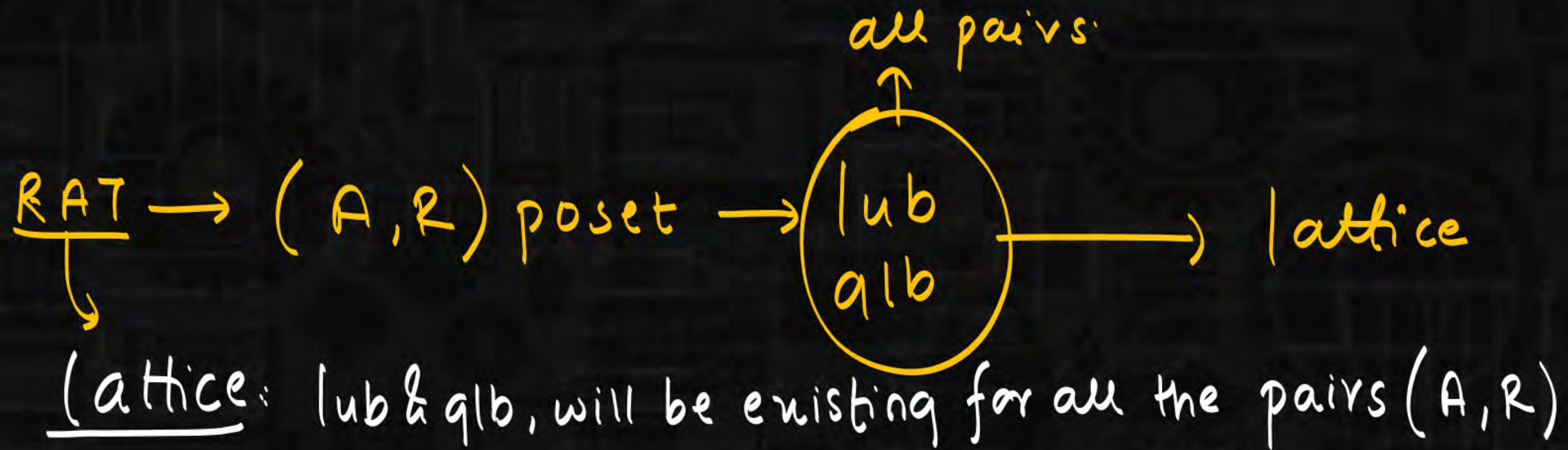
$$\text{glb}(b, c) = a$$

$$\text{lub}(b, c) =$$

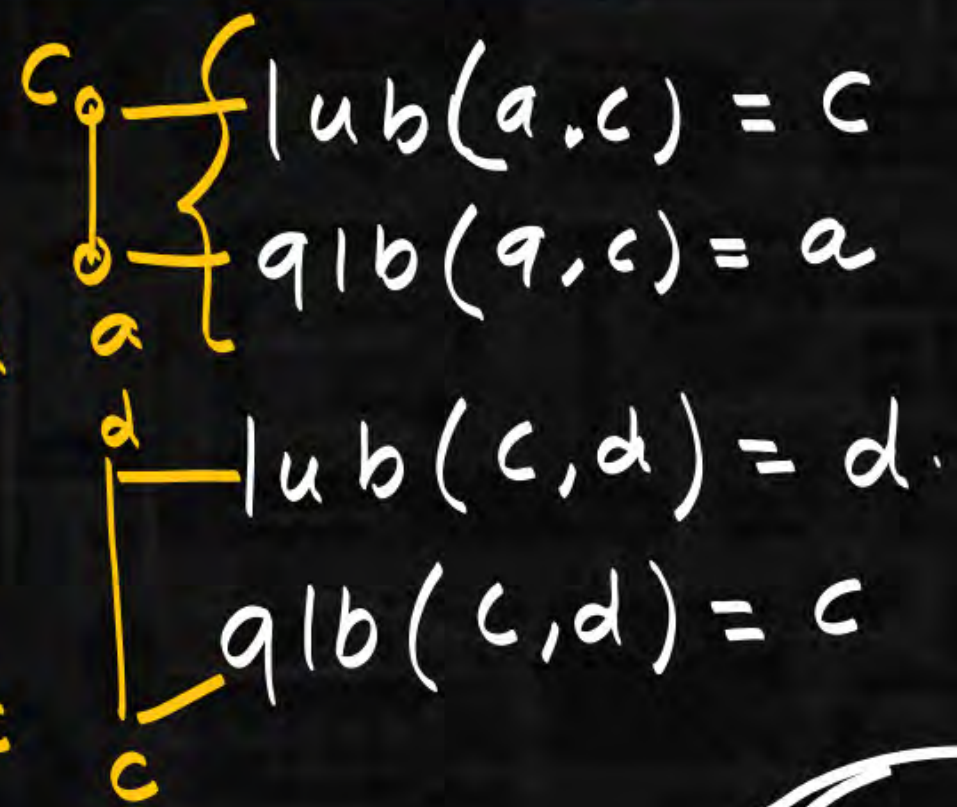
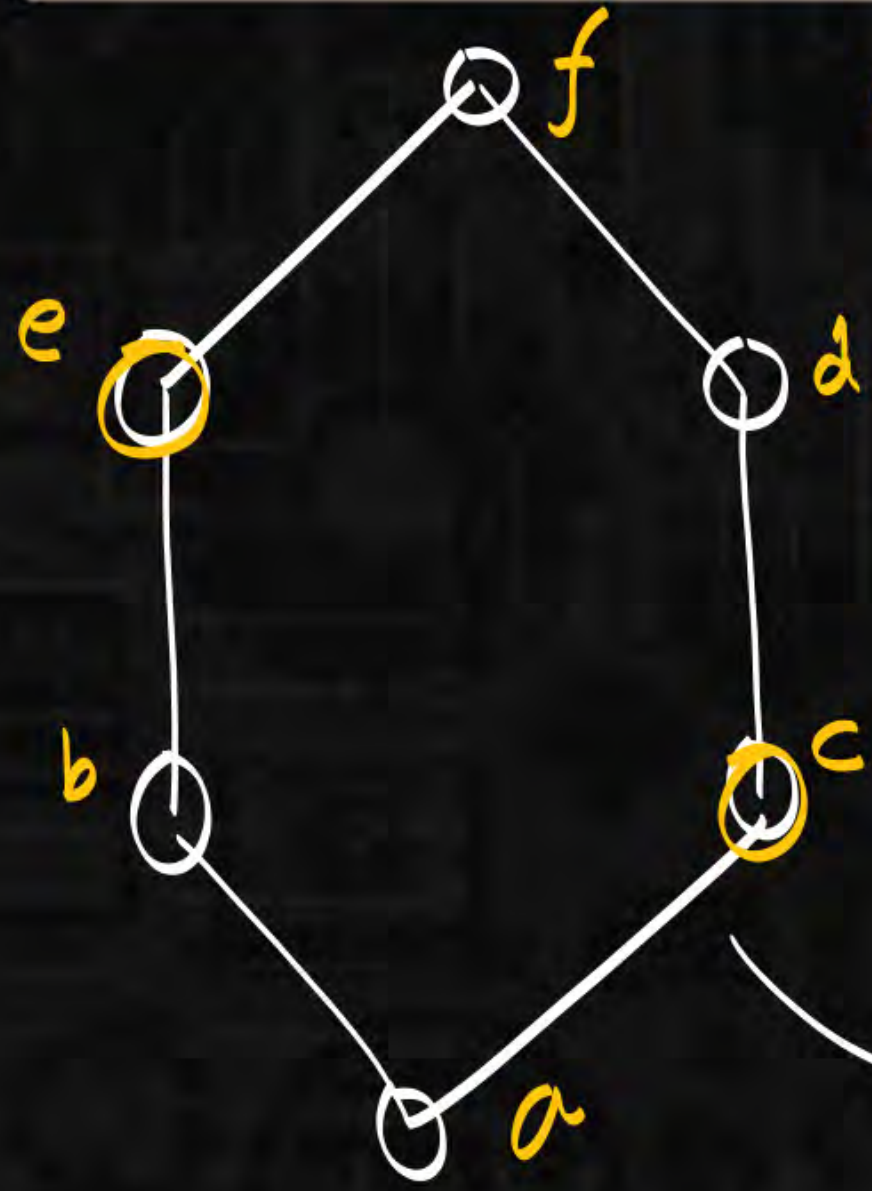
$$\text{ub}(b, c) = \{e, d, f\}$$

$$\text{lub}(b, c) = \text{N.A.}$$

Relations



Relations



$$\begin{cases} \text{lub}(e, c) = f \\ \text{glb}(e, c) = a \end{cases}$$

(A, R) lattice

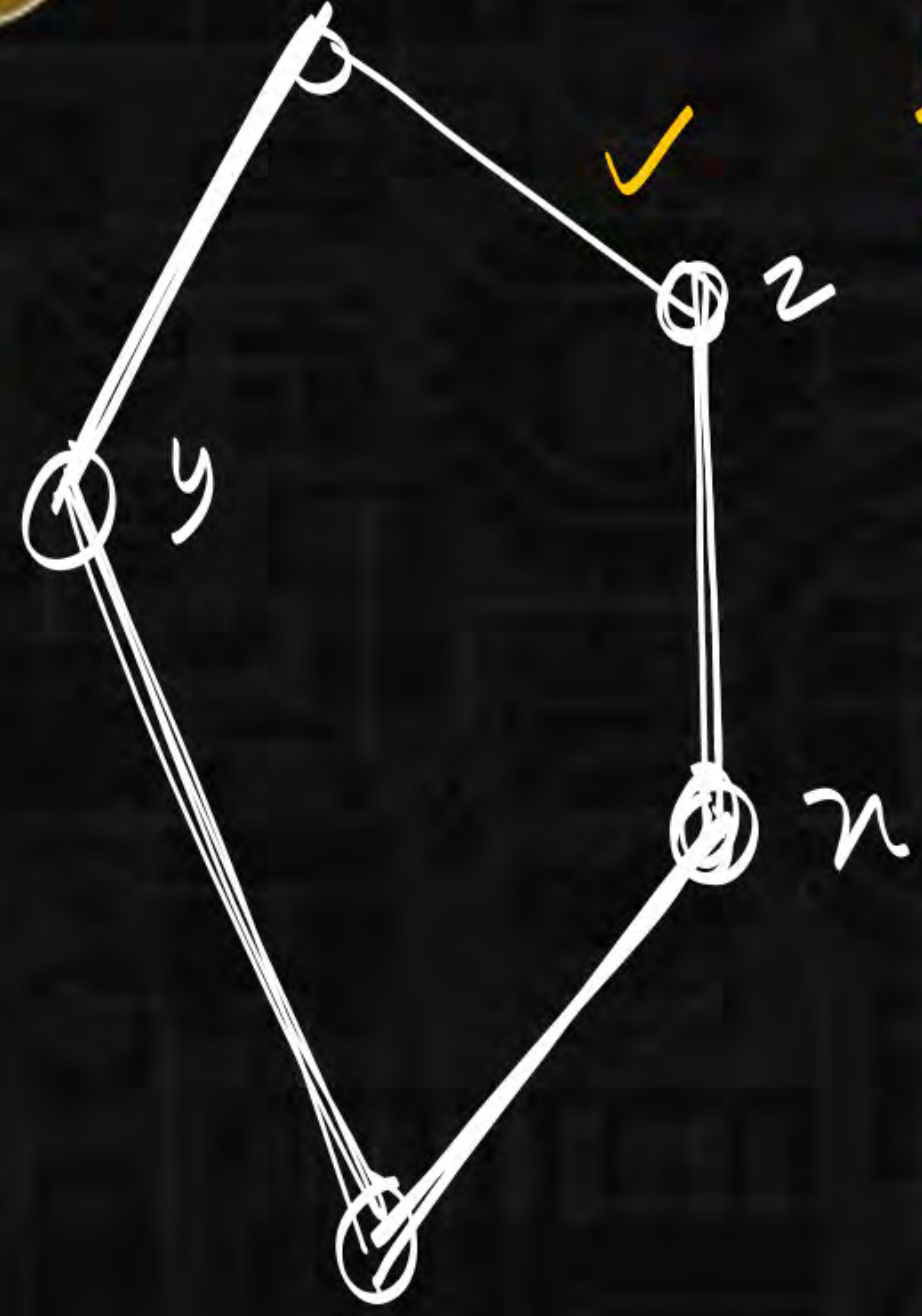
Relations

GATE

a) not lattice

$$q|b(xy)$$

$$|b(xy) = \{a, b\}$$



lattice?

Relations



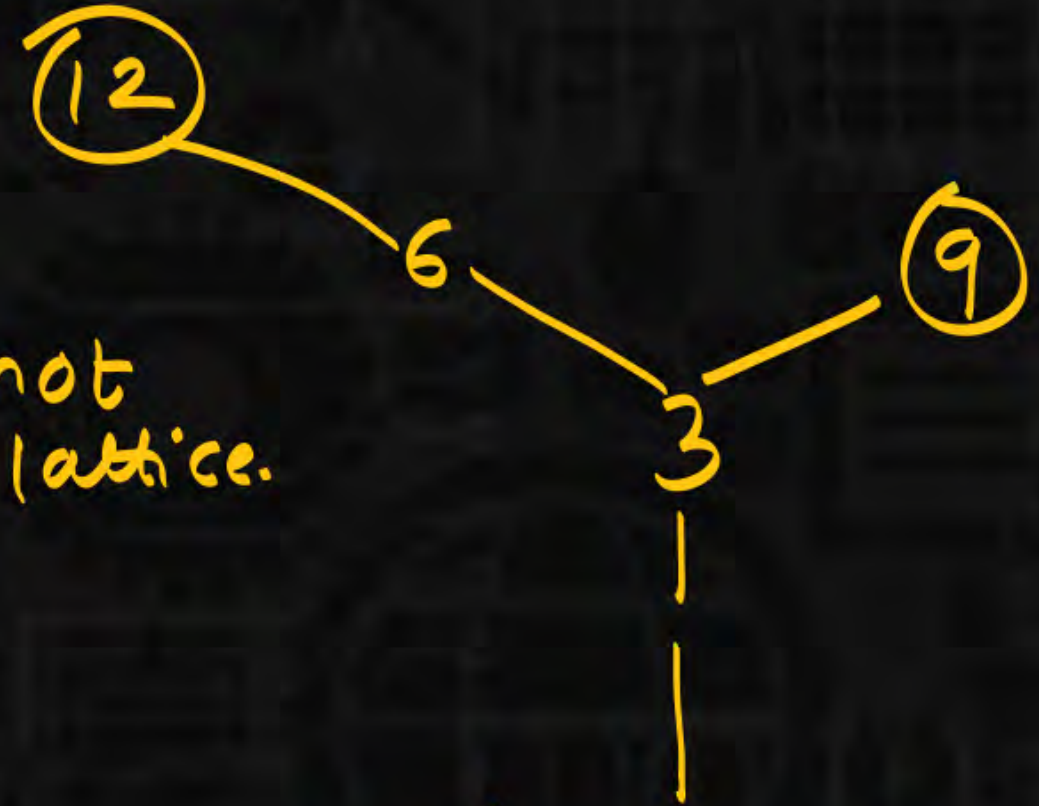
a) $\left[\{1, 3, 6, 9, 12\}, 1 \right]$

not
lattice.

b) $\left[\{1, 5, 25, 125\}, 1 \right]$

c) $\left[\{1, 2, 4, 8, 16\}, 1 \right]$

d) nota.



$$\text{glb}(12, 9) = 3.$$

$$\text{lub}(12, 9) = \text{NA}.$$

Relations

$(\{1, 2, 3, 4, 6, \dots, 10\}, \leq)$

$\left\{ \begin{array}{l} (A, R) \text{ is linear order} \\ \text{Relation then it is lattice.} \end{array} \right.$

all elements are comparable

↓
To set

↓
Chain

10
9
8
7
6
5
4
3
2
1

Total order Relation.
linear order Relation

Relations

X

(a, b)

(b, c)

(c, e)

(d, e)

(a, d)



min. no. of order pairs
that need to be added
to make (X, R)
is lattice.

Ans: 0

Relations

$$\vee \rightarrow \text{lub}$$

$$\wedge \rightarrow \text{glb}$$

(A, R) is lattice.

it satisfies following properties.

- 1) $a \vee a = a$ $a \wedge a = a$ $\text{lub}(a, a) = a$ $\text{glb}(a, a) = a$
- 2) $a \vee b = b \vee a$ $a \wedge b = b \wedge a$ $\text{lub}(a, b) = \text{lub}(b, a)$ $\text{glb}(a, b) = \text{glb}(b, a)$
- 3) $a \vee (b \vee c) = (a \vee b) \vee c$
 $a \wedge (b \wedge c) = (a \wedge b) \wedge c$
- 4) $a \vee (a \wedge b) = a$
 $a \wedge (a \vee b) = a$

Relations

$$\begin{aligned}
 2) \quad & a \vee b = b \vee a \\
 & \text{lub}(a, b) = \text{lub}(b, a) \\
 & a \wedge b = b \wedge a \\
 & \text{glb}(a, b) = \text{glb}(b, a)
 \end{aligned}$$

$$1. \left\{ \begin{aligned} & a \vee a = a. \\ & \text{lub}(a, a) = a \\ & a \wedge a = a. \\ & \text{glb}(a, a) = a. \end{aligned} \right.$$

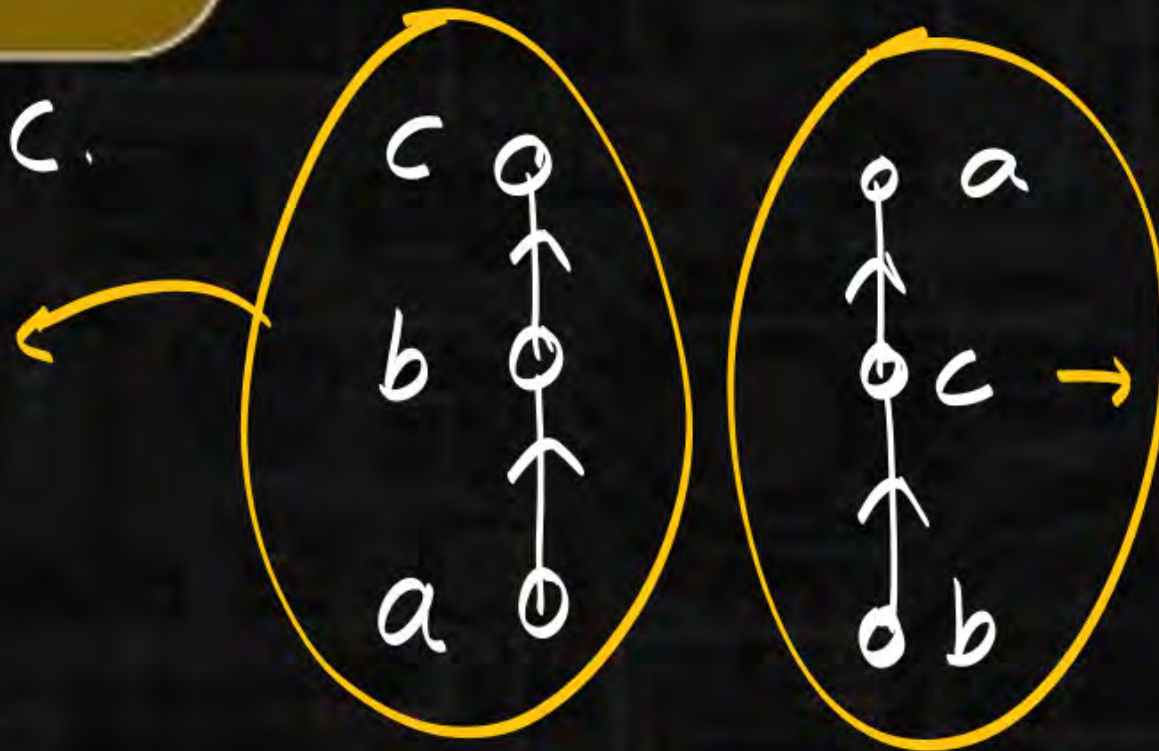
Relations



$$\vee \rightarrow \text{lub} \quad \wedge \rightarrow \text{glb}$$

$$3) i) a \vee (b \vee c) = (a \vee b) \vee c$$

$$ii) a \wedge (b \wedge c) = (a \wedge b) \wedge c$$



$$(a \vee b) \vee c$$

$$\text{lub}(a, b)$$

$$a \vee c$$

$$\text{lub}(a, c) = a$$

$$a \vee (b \vee c)$$

$$\text{lub}(b, c)$$

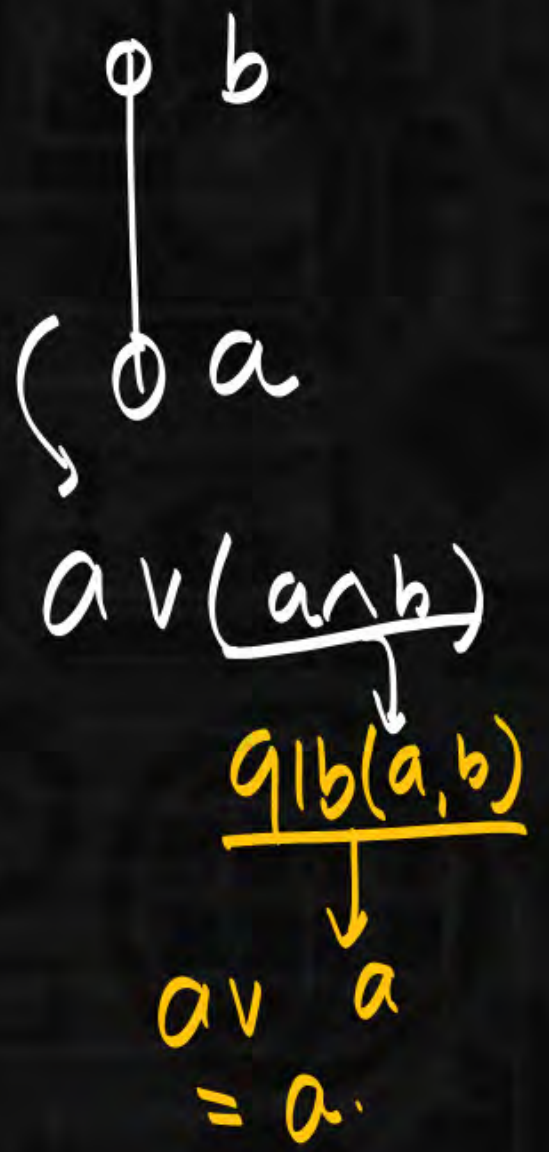
$$a \vee c$$

$$\text{lub}(a, c)$$

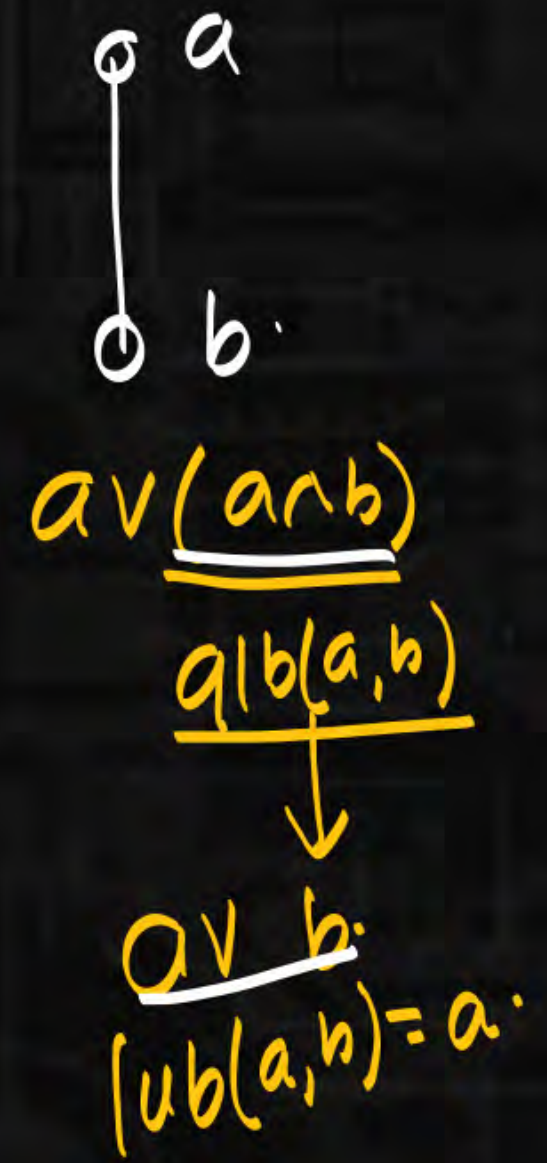
$$a$$

Relations

$$a \vee (a \wedge b) = a$$



$$a \wedge (a \vee b) = a$$



$\vee \rightarrow \text{lub}$
 $\wedge \rightarrow \text{glb}$

